

SigLIP · arXiv 2303.15343 · Sep 14

# Visionary

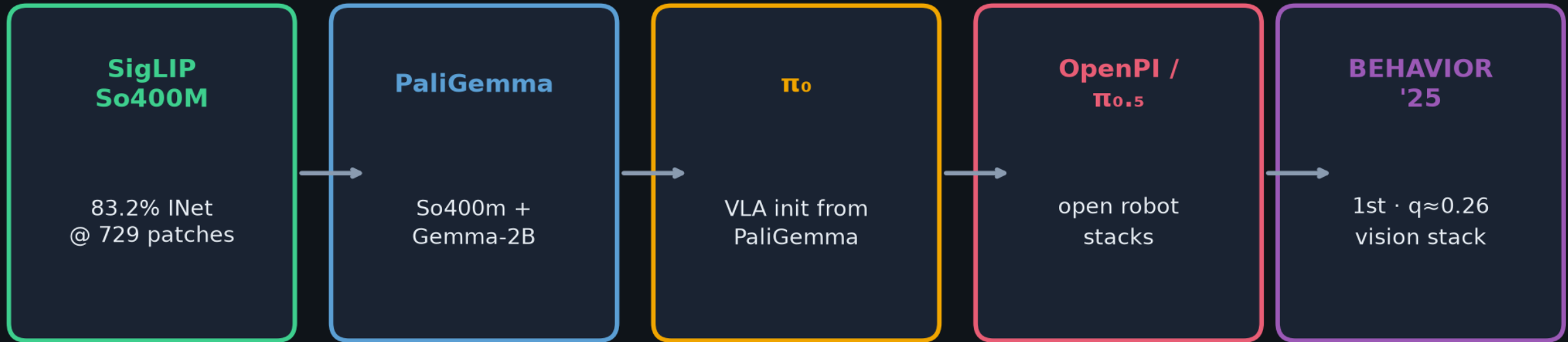
seat 2 · 11+3

2026 encoder-swap · SigLIP 2 · q-score

Paper club · VLA Architectures · dark academic

# So400M → open VLA eyes

Public efficiency point that became the default vision tower

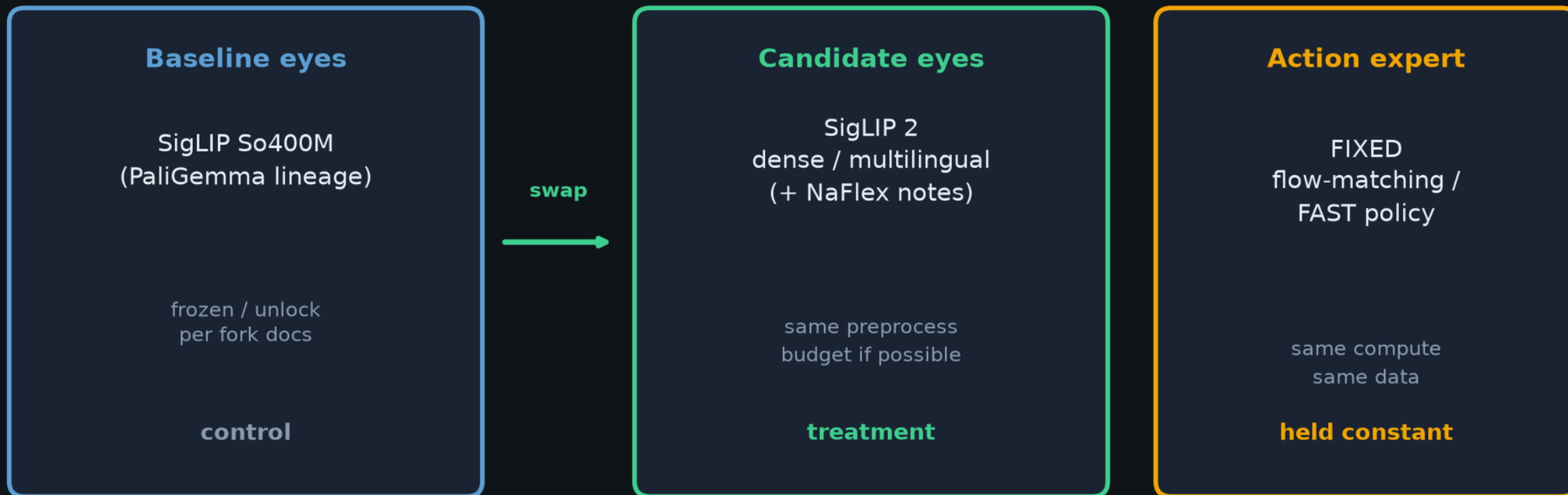


**Perception backbone — NOT a classical motion planner**

Chunking / stages / recovery / data still sit downstream of these eyes

# 2026 BEHAVIOR · encoder-swap experiment

Swap eyes · keep the same action expert · measure q-score



**Primary metric: q-score under fixed policy compute — NOT COCO R@1**

# Ideas enabled by pairwise loss

## Async / hard-neg

Small robot batches:  
mine hard negatives  
without full-batch  
softmax sync

## NaFlex / resolution

Egocentric cams &  
wrist fisheye —  
flexible resolution  
without square crop tax

## Dense / multilingual

SigLIP 2 path:  
better localization &  
non-English task  
strings

## Default open eyes

Treat SigLIP→PaliGemma→ $\pi_{0.5}$   
as baseline; differentiate elsewhere

## Measure what matters

Encoder swaps in q-score  
under fixed policy compute

## 2026 research agenda · perception only

A1 SigLIP 2 behind same  $\pi_{0.5}$  action expert — q-score delta

A2 Token-length mismatch audit (16-64 vs long task strings)

A3 Hard-neg mining under sigmoid for teleop/robot captions

A4 NaFlex / multi-res for head + wrist views

A5 Freeze vs unlock ablation — document every fork

A6 Don't claim encoder swap “solves” long-horizon control

## Club thread · long-horizon control

### SigLIP IS

- Perception / vision tower
- Default open eyes for VLAs
- Standardized backbone
- Image-text pretrain recipe

### SigLIP is NOT

- A motion planner / PDDL
- A substitute for chunking
- A recovery / System-2 loop
- Proof that sigmoid loss = VLA win

**BEHAVIOR failures more often sit in chunking / stages / recovery / data**

than in ImageNet-0 of the vision tower — but So400m is still the default open eyes those controllers see through.

## One-sentence lineage

**CLIP / ALIGN / LiT + WebLI**  
**→ SigLIP → PaliGemma →  $\pi_0$  / OpenPI**  
**→  $\pi_{0.5}$  → 2025 BEHAVIOR winners**  
**(and → SigLIP 2)**

### Honest hedges

Freeze/unlock of SigLIP weights in every OpenPI fork is not always documented.  
“Uses SigLIP”  $\neq$  isolating the sigmoid loss vs data / architecture / scale.

# Don't overclaim

Encoder swap is a measured bet · not a solved household robot



# Visionary close

SigLIP 2 swap · q-score · async/hard-neg · NaFlex